

# EL: Trend-to-Product Ranking with Bayesian HIL Calibration

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**Abstract**—We cast trend-driven dropshipping as a constrained daily ranking problem over noisy Indian trend streams. EL is a deployed 70-node n8n workflow that instantiates this problem through deterministic topic scoring, overlap-based deduplication, LLM-mediated contraction, and Bayesian recalibration from Telegram feedback. The front-end score is additive and inspectable; the feedback layer is a closed-form Beta–Bernoulli shrinkage rule with constant-time updates. On the 2026-04-26 production trace, EL produced 10 curated opportunities and 30 persisted product rows. Under a controlled synthetic-feedback evaluation on this trace, calibration improves mean Spearman correlation from 0.45 to 0.85 and reduces mean Brier score from 0.335 to 0.243. The contribution is a mathematically explicit operational pipeline rather than an opaque model family.

**Index Terms**—ranking, Bayesian calibration, workflow automation, human-in-the-loop, e-commerce.

## I. PROBLEM STATEMENT

Let  $T_d = \{t_i\}_{i=1}^{n_d}$  denote the day- $d$  topic multiset collected from heterogeneous public sources, and let  $P_d$  denote the induced pool of supplier or marketplace products reachable from those topics. The target object is not a trend list but a bounded operator-review set

$$\Pi_d^* = \arg \max_{\Pi \subseteq P_d} \sum_{p \in \Pi} U(p) \quad \text{s.t.} \quad |\Pi| \leq K, A\Pi \leq b, \quad (1)$$

where  $U(p)$  is latent review utility,  $K$  is the daily queue budget, and  $A\Pi \leq b$  encodes hard topic and provider caps. Because  $U$  is unobserved,  $t \mapsto p$  is many-to-many, and external APIs are volatile, EL adopts a hybrid decomposition into deterministic filtering, tool-augmented contraction, and posterior correction rather than a single learned predictor [3], [4], [6].

## II. RANKING PIPELINE

**Phase 1: deterministic trend scoring.** For topic  $t$ , let  $h(t)$  be the lowercase concatenation of title and related-query hints. EL assigns

$$s(t) = \text{clip}_{[0,1]}(0.30 m_{T_1}(t) + 0.15 m_{T_2}(t) + 0.10 m_{T_3}(t) - 0.20 m_N(t)), \quad (2)$$

where  $m_S(t) = \sum_{k \in S} \mathbf{1}[k \in h(t)]$  counts lexicon hits,  $T_1, T_2, T_3$  are transactional, comparative, and product-context lexicons, and  $N$  is a negative-intent lexicon. The score is non-parametric, reproducible, and cost-free at

inference time. A parallel dictionary map emits one or more coarse product categories from a fixed 13-class taxonomy.

**Deduplication.** With stopword-filtered token set  $w(t)$ , EL merges  $t_i, t_j$  whenever  $|w(t_i) \cap w(t_j)| / \min\{|w(t_i)|, |w(t_j)|\} > 0.70$ , retaining the representative with richer side context.

**Opportunity contraction and enrichment.** The top-30 topics are passed to a Gemini 2.5 Flash agent with Tavily retrieval and Postgres memory. The agent emits at most 10 opportunities with raw score  $r \in [0, 10]$ . Each opportunity then fans into a structured CJ branch and a marketplace branch with Tavily search, Browserbase fallback, and Gemini extraction for content-thin pages [8]. Both branches normalize to a common review schema.

**Constrained queue construction.** Selection is greedy under explicit caps: minimum score, per-topic cap, per-provider cap, and total queue cap. The deployed main workflow contains 70 nodes, of which 60 are functional; a separate 3-node error workflow handles uncaught execution failures.

## III. FEEDBACK AND EVIDENCE

Telegram callbacks are interpreted as Bernoulli supervision rather than terminal moderation. For category  $c \in \mathcal{C}$  and callback  $y \in \{0, 1\}$  (*approve*=1, *reject*=0; *skip* ignored), EL maintains  $\theta_c \mid \mathcal{D}_c \sim \text{Beta}(\alpha_c, \beta_c)$  with update  $(\alpha_c, \beta_c) \leftarrow (\alpha_c + y, \beta_c + 1 - y)$ . With posterior mean  $\hat{\mu}_c = \alpha_c / (\alpha_c + \beta_c)$  and effective sample size  $n_c = (\alpha_c + \beta_c) - 2$ , the calibrated score is

$$\text{cal}(p) = w_c \hat{\mu}_c + (1 - w_c) \frac{r(p)}{10}, \quad w_c = \frac{n_c}{n_c + n_0}, \quad n_0 = 5. \quad (3)$$

Hence cold start yields identity calibration, while repeated feedback shifts ranking mass toward category-specific empirical approval rates. The update is conjugate, exact, and constant-time [10].

The documented production execution on 2026-04-26 produced 10 curated opportunities and 30 persisted product rows. Raw opportunity scores ranged from 7.5 to 8.8 with mean 8.05. The observed persisted categories were *Pumps*, *Rings*, and *Blouses & Shirts*. Because a sufficiently large real callback corpus does not yet exist, the feedback layer was validated under a controlled synthetic model defined by  $\Pr(Y = 1 \mid r, c) = \sigma(0.8(r - 8) + b_c)$  with  $b = (-1.5, +1.5, 0)$ . Across 1000 bootstrap train/test splits (15/15), the raw score  $r/10$  achieved mean Spearman  $\rho = 0.446$  and mean

Brier score 0.335, whereas Equation (3) achieved  $\rho = 0.853$  and Brier 0.243 [12]. The implication is not that EL has solved real-world product selection, but that when reviewer preference contains a category effect orthogonal to the agent score, a conjugate posterior can recover that effect under sparse supervision.

Three caveats delimit the claim: the trace is single-run, the calibration evidence is synthetic, and the enrichment branches remain exposed to exogenous API drift. The next empirical step is to replace this trace with a longitudinal callback corpus and compare the deterministic front-end and Bayesian rear-end against learned baselines on labeled intent data. Until then, EL is best read as a dense systems paper: mathematically explicit, operationally grounded, and deliberately narrow in its claims.

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